

21/12/22

Indian Institute of Engineering Science and Technology, Shibpur
B. Tech. 3rd Semester Final Examination, 2022
Subject: Digital Logic (CS - 2102)

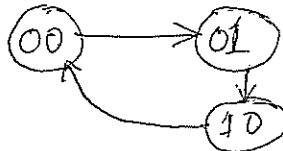
Full marks: 50

Time: 3 hours

- 1(i) Define fan out and propagation delay time.
- (ii) What is the advantage of totem-pole configuration of TTL over open-collector configuration?
- (iii) What is the advantage of using priority encoder over normal encoder?
- (iv) Define flip-flop.
- (v) Write 2-bit gray code and 2-bit binary code.
- (vi) Show the addition of two decimal number, 7 and 4.
- (vii) How many clock pulses can be counted using (a) 3-bit ring counter (b) 3-bit Johnson counter
- (viii) Write the state table of a positive edge triggered flip-flop whose S input is connected to Q and R input is connected to Q, initial value of Q is 1 and Q' is 0.
- (ix) What do you mean by self-correcting counter?
- (x) Draw the block diagram of 1-bit full adder using half-adder and logic gate.
- (xi) What is the role of a parity bit in data transmission?
- (xii) Simplify the Boolean function, $F(x,y,z)=(x+y+z)(x+y+z')(x+y'+z)(x+y'+z')$
- (xiii) Simplify the Boolean function, $F(A,B,C) = A'B'C' + B'CD' + A'BCD' + AB'C'$
- (xiv) Give two examples of non-weighted code.
- (xv) What is the (84-2-1) representation of BCD 1001? [2x15=30]

2. Answer any two from the following questions

- (i) Show the state diagrams of (a) S-R flip-flop (b) D flip-flop (c) J-K flip-flop (d) T flip-flop [4x2]
- (ii) Write the truth table of BCD to excess-3 code converter, derive the Boolean equation of the output signals of the converter and draw the logic circuit [2+4+2]
- (iii) Write the count sequence of BCD ripple counter, show the state diagram, explain the design details of the counter and draw the logic circuit of the counter [1+1+4+2]
- (iv) The state diagram of a 2-bit synchronous counter is given below.



Write the excitation table to design the counter using J-K flip-flop, explain the design details of the counter and draw the logic circuit of the counter using J-K flip-flop [2+4+2]

3. Answer any one from the following questions

- (i) Draw the following circuit using open-collector TTL inverter and write the truth table.
 (a) wired-AND (b) wired-OR [2x2]
- (ii) Draw the circuits of 3-bit left shift and 3-bit right shift register using D flip-flop [2x2]
- (iii) Compare the performance of a 4-bit ripple carry adder and a 4-bit carry look ahead adder in terms of propagation delay and hardware complexity [4]